

British Math Olympiad Rd 1 2001 — P5/5

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A triangle has sides of length a, b, c and its circumcircle has radius R . Prove that the triangle is right-angled if and only if $a^2 + b^2 + c^2 = 8R^2$.

Solution

Let $\triangle ABC$ have side lengths $a \leq b \leq c$ with the normal labelling convention.

First we show that if $\triangle ABC$ is right-angled then $a^2 + b^2 + c^2 = 8R^2$. Observe that side AB is a diameter of the circumcircle. This follows from the converse of the "angles in a semi circle are 180° " circle theorem. Thus $c = 2R$. Now by Pythagoras,

$$a^2 + b^2 + c^2 = 2c^2 = 2(4R^2) = 8R^2.$$

Now we show that if $a^2 + b^2 + c^2 = 8R^2$ then $\triangle ABC$ is right-angled. Note that we must have $c \leq 2R$ because otherwise the triangle cannot be inscribed within its circumcircle. By cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos \angle C \leq 4R^2$$

Now substituting $a^2 + b^2 = 8R^2 - c^2$ yields

$$(8R^2 - c^2) - 2ab \cos \angle C \leq 4R^2$$

$$4R^2 \leq c^2 + 2ab \cos \angle C$$

$$0 \leq 4R^2 - c^2 \leq 2ab \cos \angle C$$

where the third line is because $c^2 \leq 4R^2$. So this tells us that $\cos \angle C \geq 0$. Thus from the cosine rule formula we must have

$$c^2 \geq a^2 + b^2.$$

Now observe that

$$8R^2 = a^2 + b^2 + c^2 \leq (c^2) + c^2 \leq 8R^2.$$

But this equality case can only happen if $c^2 = a^2 + b^2$. That is $\triangle ABC$ is right-angled with hypotenuse c . This completes the proof. ■